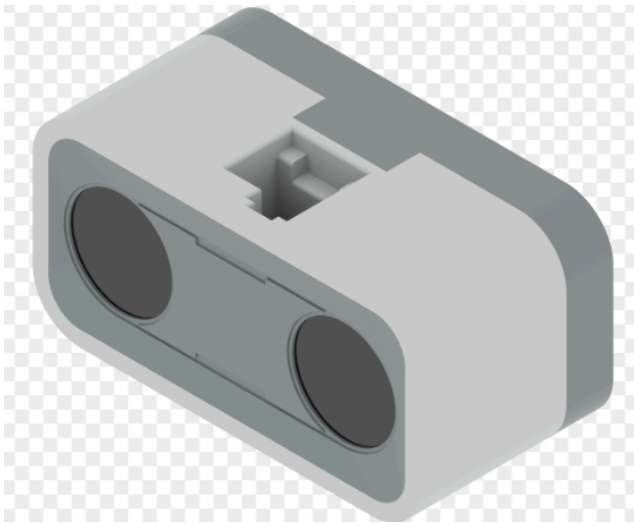


Bumper & Distance Sensor

1. What Do They Look Like?

Distance Sensor

- **Appearance:** Has two circular "eyes" in front (like a pair of binoculars) Gen 1.
- **Ports:** Connects to a *Smart Port*.
- **Example Image:**



And a Gen 2



Distance Sensor Uses

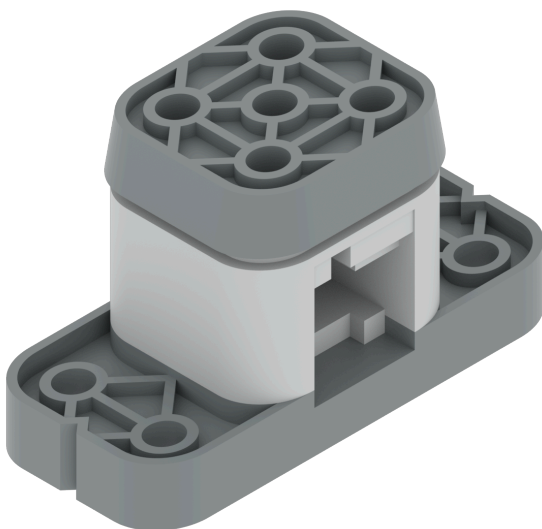
- Measures how far an object is in **millimeters (mm)** or **inches (in)**.
- Uses infrared light to detect objects.
- Can detect if an object is near/far or follow an object.

Use Cases:

- Avoiding obstacles
- Following a wall or object
- Keeping distance from another robot

Bumper Switch

- **Appearance:** Small, black, and has a round red button on top.
- **Ports:** Connects to a *Smart Port*.
- **Example Image:**



Bumper Switch Uses

- Detects if the button is pressed or released (True or False).
- Very simple, reliable input sensor.

Use Cases:

- Starting or stopping actions
- Detecting collisions
- Controlling turns or arm movements

3. How to Connect Sensors in VEXcode IQ

1. Plug the sensor into any *Smart Port* on the VEX IQ Brain.
2. Open **VEXcode IQ (Blocks)**.
3. Click the **Devices** icon (top right gear).
4. Click **Add a Device**.
5. Select **Distance Sensor** or **Bumper Switch**.
6. Choose the correct port (the one you plugged it into).
7. Click **Done**.